

Optimization

6.1 The multiobjective problem

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Until now, “better” was clear

Chapter 6. Multiobjective optimization

Throughout the course we have solved

$$\min_{\mathbf{x} \in X} f(\mathbf{x}).$$

Given two feasible solutions $\mathbf{x}^A$ and $\mathbf{x}^B$,

$$f(\mathbf{x}^A) < f(\mathbf{x}^B)$$

simply means:

$\mathbf{x}^A$ is better than $\mathbf{x}^B$.

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simply means:

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What if there are two things we
want to improve at the same time?

(This is the question that closed 5.3.)

A problem everyone understands

From Nantes to Lille. I care about the **duration** and the **cost**.

option	duration	cost
Car via Rouen	5h49	251 €
TGV	4h02	84 €
Plane	1h05	134 €

Which would you choose?

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Car vs. TGV: $4\text{h}02 < 5\text{h}49$ and $84 < 251$.

The TGV is both faster and cheaper: the car via Rouen makes no sense as a choice.

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TGV vs. plane: the plane is faster; the TGV is cheaper. **Which is better?**

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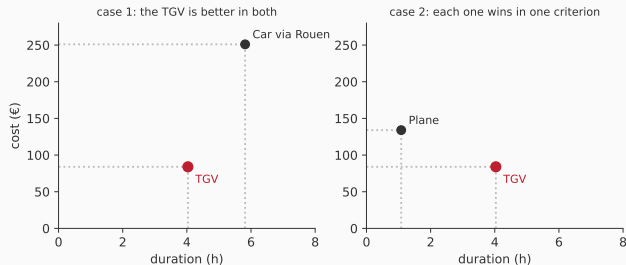
Car vs. TGV: $4h02 < 5h49$ and $84 < 251$.

The TGV is both faster and cheaper: the car via Rouen makes no sense as a choice.

TGV vs. plane: the plane is faster; the TGV is cheaper. **Which is better?**

We can no longer answer without knowing how much an hour is worth to us, compared with a euro. **This is where the multiobjective problem is born.**

There are two kinds of comparison



Case 1. Car = (5h49, 251 €), TGV = (4h02, 84 €).

The TGV is better in both criteria:

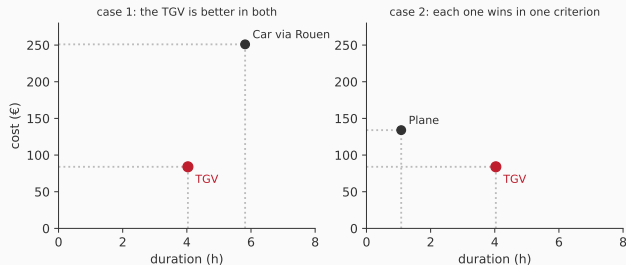
the car can be eliminated.

Case 2. TGV = (4h02, 84 €), plane = (1h05, 134 €).

The plane wins on time, the TGV on cost:

neither is unequivocally better.

There are two kinds of comparison



Case 1. Car = (5h49, 251 €), TGV = (4h02, 84 €).

The TGV is better in both criteria:

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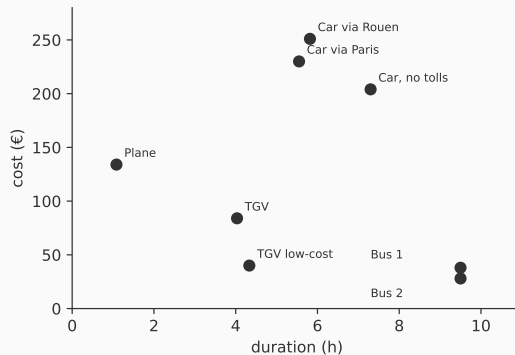
neither is unequivocally better.

What are these two situations called, mathematically? (Answer in 6.2.)

Now, all the options

option	duration	cost (€)
Car via Paris	5h33	230
Car via Rouen	5h49	251
Car, no tolls	7h18	204
Bus 1	9h30	38
Bus 2	9h30	28
TGV	4h02	84
TGV low-cost	4h20	40
Plane	1h05	134

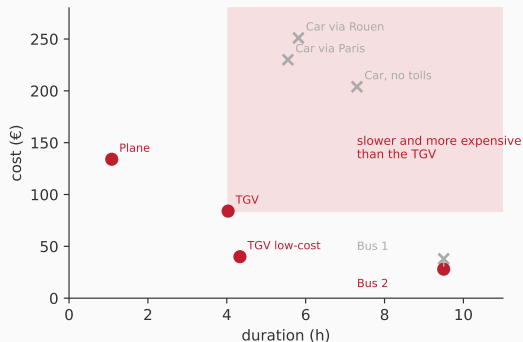
Eight ways of getting from Nantes to Lille. Which can we eliminate?



Now, all the options

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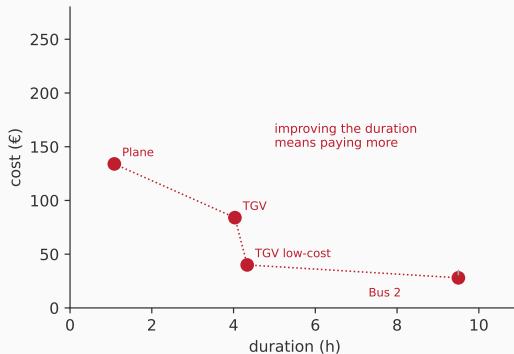
Out go the three cars (worse than the TGV in both criteria) and bus 1 (same duration as bus 2, and more expensive).



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Four remain. Among them, improving one criterion worsens the other.



The multiobjective problem does not, in general, end in a single solution: it ends in a **set of trade-offs**.

So, which one do we choose?

Someone says: “for me, an hour is worth 30 €”.

Then each option gets a single number,

$$\text{cost} + 30 \times \text{duration},$$

and we pick the smallest. With 10 € per hour the choice changes:

	30 €/h	10 €/h
Bus 2	313	123
TGV low-cost	170	83
TGV	205	124
Plane	167	145

So, which one do we choose?

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But the “30 €/h” is a **preference** — it does not come from the problem.

Without additional preferences, the multiobjective model lets us **eliminate** the clearly worse options, but it **does not establish a total order** among the trade-offs.

Optimization identifies the trade-offs.

The **decision-maker** chooses among them.

From travel to engineering

This is not a travel problem. It is what happens in engineering:

weight	↔	stiffness
cost	↔	performance
energy	↔	comfort
efficiency	↔	emissions

In engineering, the interesting objectives are often conflicting: reducing the mass may conflict with increasing the stiffness.

Viscoelastic sandwich panels (IDMEC)

Maximize the **damping** η ; minimize the **mass** and the **cost**. Three of the designs obtained:

design	η	mass	cost
3636	40.7%	1.49 kg	346 USD
5881	13.0%	0.41 kg	—
2701	4.7%	—	21 USD

The one with the most damping is the heaviest and the most expensive. Which is the best?

Madeira, Araújo, Mota Soares, Mota Soares, Ferreira, Compos. Part B 77:391–401, 2015 (— : value not given in the text).

Now for real: the mathematical problem

$$\min_{\mathbf{x} \in X} \mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_m(\mathbf{x})), \quad X = \{\mathbf{x} \in \mathbb{R}^n : g_j(\mathbf{x}) \geq 0, h_\ell(\mathbf{x}) = 0\}$$

$\mathbf{x}$ the **decision** we take (a design, a trip)

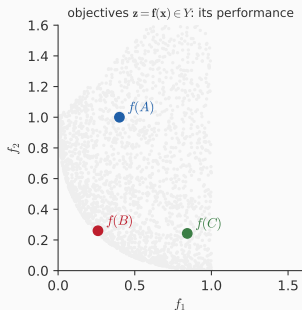
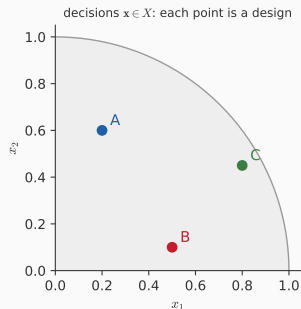
X the **feasible** decisions

$f_1, \dots, f_m$ the **criteria** we want to minimize, $m \geq 2$

$\mathbf{f}(\mathbf{x})$ the **performance** of that decision in all criteria — a vector

Maximizing f_i is the same as minimizing $-f_i$.

There are two different spaces



Decision space: $\mathbf{x} \in X \subset \mathbb{R}^n$. Each point is
a **solution** (a design).

Objective space: $\mathbf{z} = \mathbf{f}(\mathbf{x}) \in Y \subset \mathbb{R}^m$. Each
point is the **performance** of that design.

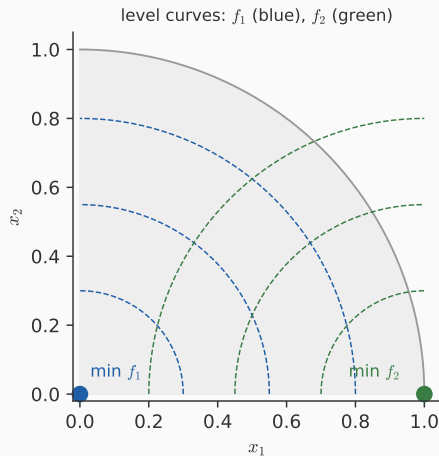
$\mathbf{x} \neq \mathbf{f}(\mathbf{x})$: one is the solution; the other is its objectives.

E.g.: design $B = (0.5, 0.1)$ has performance $\mathbf{f}(B) = (0.26, 0.26)$.

The convex example of the chapter

$$\min \begin{cases} f_1(\mathbf{x}) = x_1^2 + x_2^2 \\ f_2(\mathbf{x}) = (x_1 - 1)^2 + x_2^2 \end{cases}$$

with $\mathbf{x} \in [0, 1]^2$, $x_1^2 + x_2^2 \leq 1$.

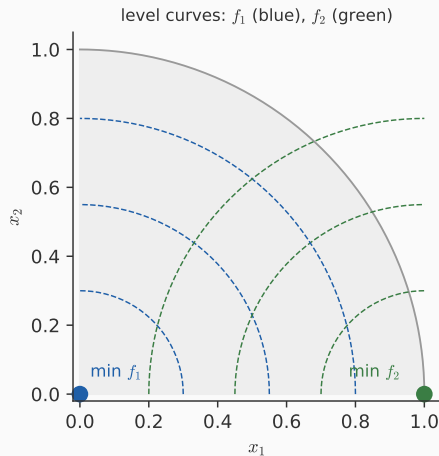


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Where is the minimum of f_1 ? At $(0, 0)$. And that of f_2 ? At $(1, 0)$.



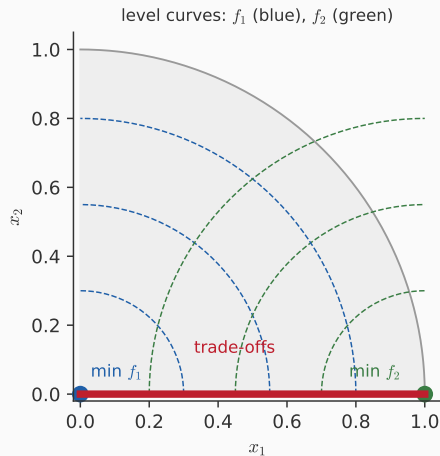
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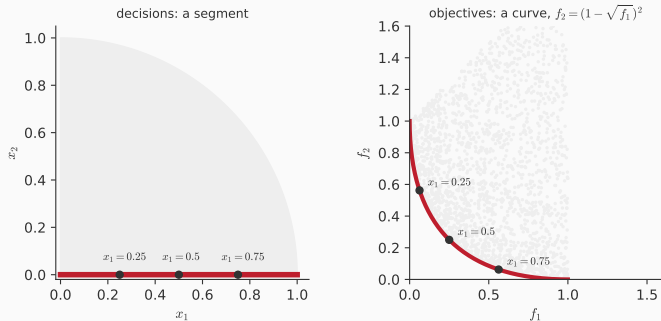
with $\mathbf{x} \in [0, 1]^2$, $x_1^2 + x_2^2 \leq 1$.

Where is the minimum of f_1 ? At $(0, 0)$. And that of f_2 ? At $(1, 0)$.

What if we want to improve both? Off the segment between $(0, 0)$ and $(1, 0)$, lowering x_2 to 0 improves f_1 and f_2 . On the segment, moving towards one minimum moves away from the other: these are the **trade-offs**.



The same solution seen in both spaces



On the left, the trade-offs: the segment $x_2 = 0$, $0 \leq x_1 \leq 1$. On the right, their images: the curve $f_2 = (1 - \sqrt{f_1})^2$.

A simple set in the decision space can become a curve in the objective space. The decision-maker looks at the right (what is gained and what is lost); the design to be built is on the left.

Three ways of dealing with several objectives

A priori

I say **beforehand** what I prefer: “an hour is worth 30 €”, weights, limits (“the cheapest as long as it takes less than 5 hours”).

Optimization returns one solution.

A posteriori

First I **see the trade-offs**; I choose afterwards.

Optimization returns a set.

Interactive

Optimization and decision **alternate**: I see some trade-offs, adjust the preferences, repeat.

In this course: mainly an *a posteriori* perspective — we want to show the decision-maker the trade-offs.

First: what is a good solution?

6.2 Pareto front

Then: how do we compute it?

One point at a time

6.3 Aggregation

6.4 ε -constraint

one scalar optimization per point, with the methods
of Chs. 2–5

Many points in one run

6.5 DMS

6.6 NSGA-II

a list or a population of points

optimization

→

trade-offs

→

decision

KEY TAKEAWAYS — 6.1

1. With several objectives, “better” usually no longer has a single answer.
2. Some solutions are clearly worse in all criteria and can be **eliminated**.
3. Among the solutions not eliminated, **trade-offs** arise: improving one objective may force another to get worse.
4. Do not confuse $\mathbf{x} \in X$ (the solution) with $\mathbf{f}(\mathbf{x}) \in Y$ (the performance of the solution).
5. $\text{optimization} \rightarrow \text{trade-offs} \rightarrow \text{decision}$.

How do we define mathematically that one solution is unequivocally worse than another?



6.2 Pareto front

References Full list: *Course bibliography*, at the end of the course PDF.

- Miettinen, K., *Nonlinear Multiobjective Optimization*, Kluwer Academic Publishers, 1999.
- Madeira, J. F. A., Araújo, A. L., Mota Soares, C. M., Mota Soares, C. A., Ferreira, A. J. M., *Multiobjective design of viscoelastic laminated composite sandwich panels*, Composites Part B: Engineering 77:391–401, 2015 — the sandwich-panel case.

Resources: —